

IoT systems

- IoT system applications
- IoT system architectures

IoT system applications

- Soft real-time networked embedded system.
 - Input devices: tags, sensors, etc.
 - Output devices: motor controllers, displays, etc.
- Examples:
 - Computer-readable identification code for objects.
 - Appliances controlled by cell phone interface.
 - Sensor network with analytics.

Devices

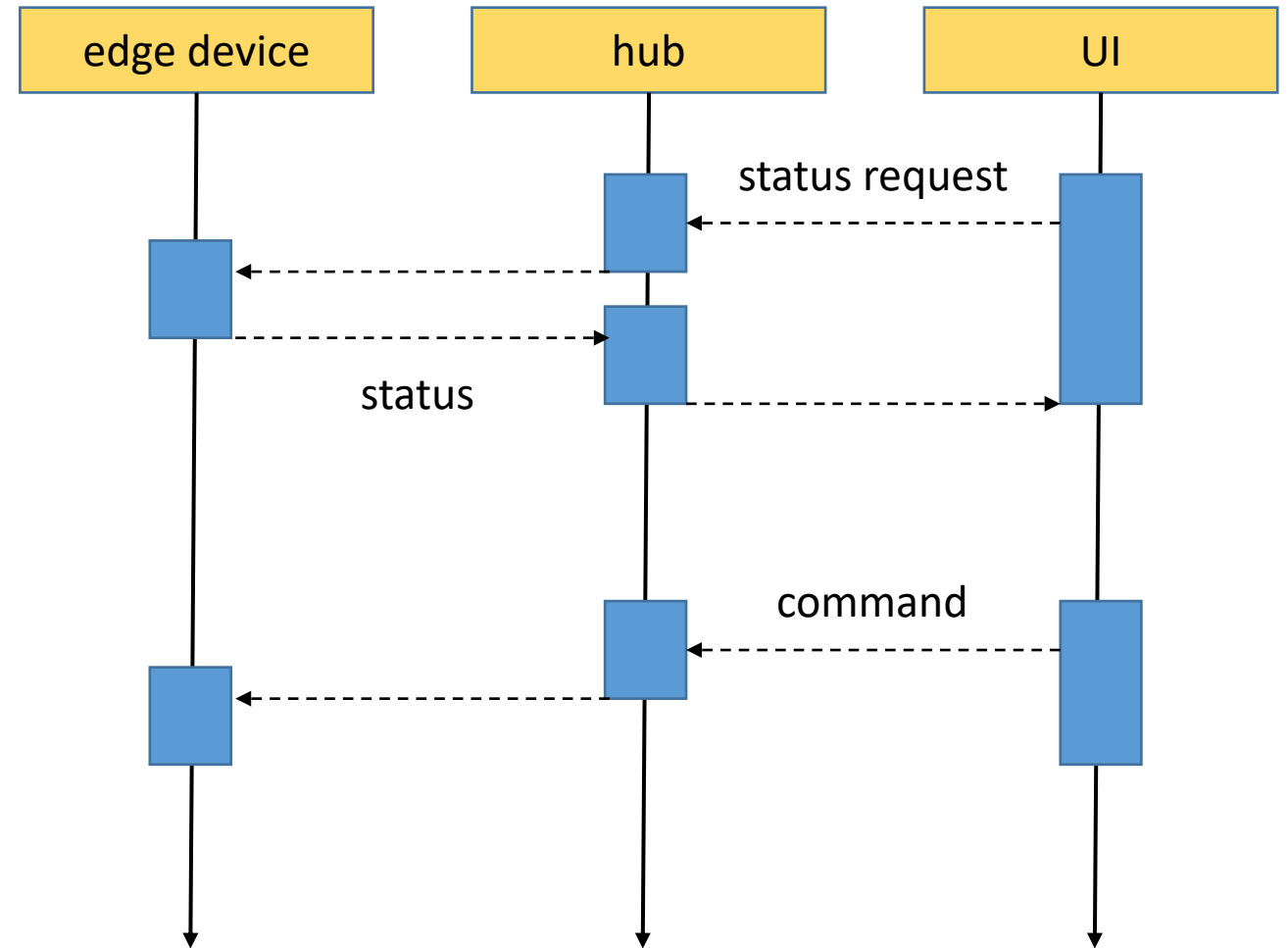
- People:
 - Implanted devices in the body.
 - Wearable devices on the body.
 - Environmental devices outside the body.
- Objects:
 - Interior: temperature sensor, etc.
 - Exterior: RFID, etc.
 - Environmental: camera, motion sensor, etc.

RFID

- RFID tag can provide object ID (Electronic Product Code, etc.), other information.
- Many tags are read-only, some are writable.
- Two types of tags:
 - Passive transmits only when it receives a request.
 - Active tag both transmits independently and responds to requests.
- Passive may also be used to refer to tags with no internal power source.
- RFID tags may operate in several different bands and at different ranges.

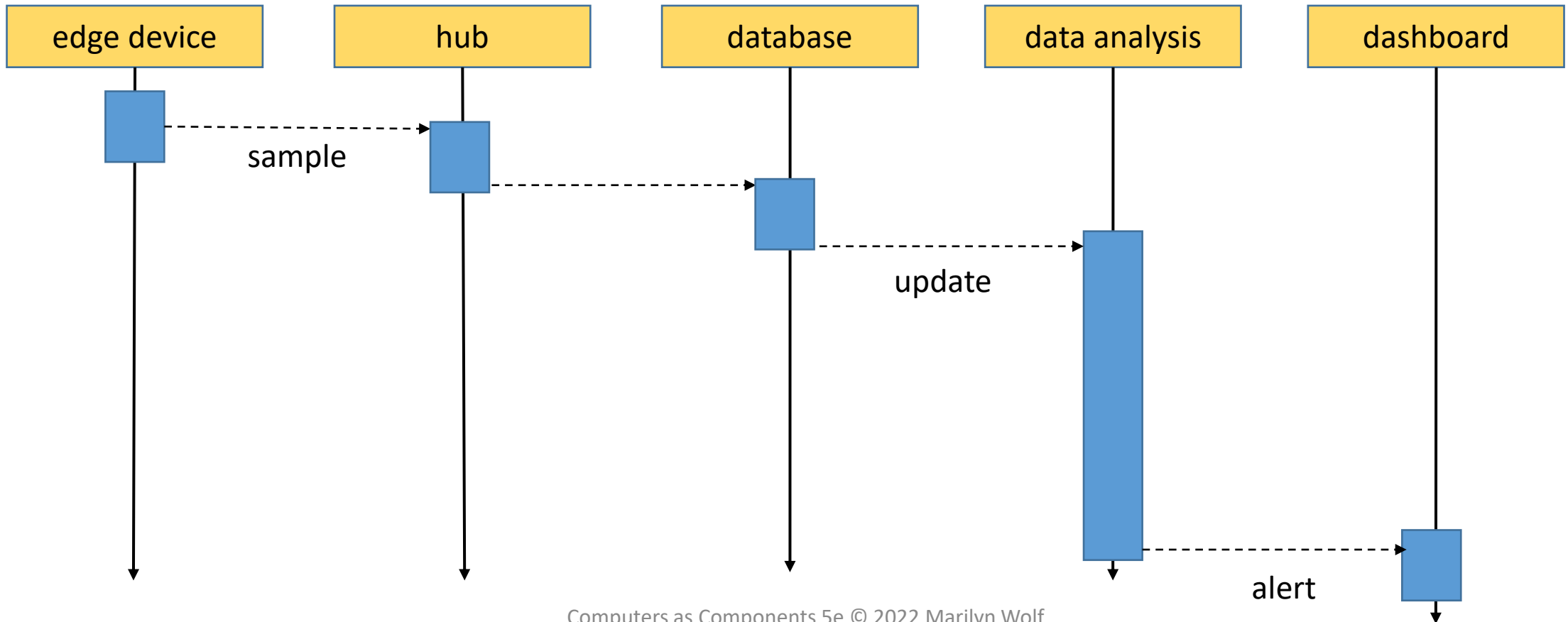
IoT system architectures

- Edge: I/O devices.
- Cloud: centralized processing.
- Smart appliance = connected appliance + network + UI.



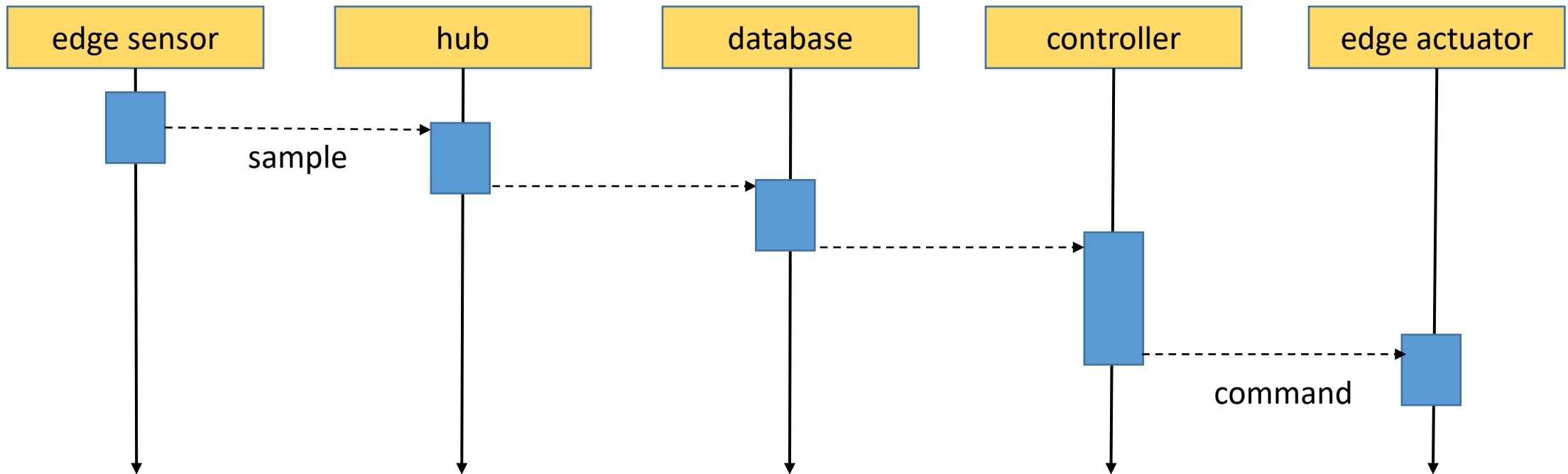
IoT system architectures, cont'd.

- Monitoring system = sensors + network + database + dashboard.



IoT system architectures, etc.

- Control system = sensors + database + controller + actuator.



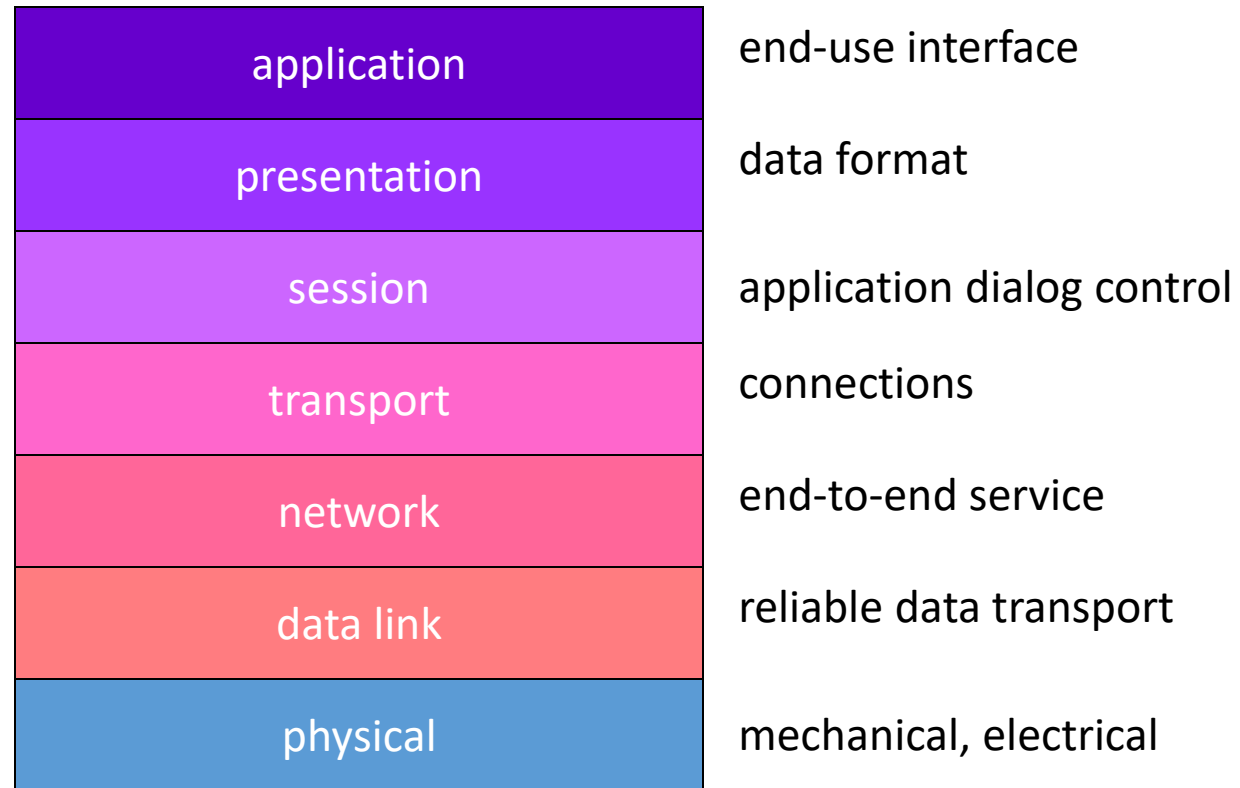
IoT systems

- OSI model for networks.
- Internet protocol.
- IoT networking concepts.
- Example networks:
 - Classic Bluetooth, Bluetooth Low Energy.
 - 802.15.4 and Zigbee.
 - Wi-Fi.

Network abstractions

- International Standards Organization (ISO) developed the **Open Systems Interconnection (OSI)** model to describe networks:
 - 7-layer model.
- Provides a standard way to classify network components and operations.

OSI model



OSI layers

- **Physical**: connectors, bit formats, etc.
- **Data link**: error detection and control across a single link (single hop).
- **Network**: end-to-end multi-hop data communication.
- **Transport**: provides connections; may optimize network resources.
- **Session**: services for end-user applications: data grouping, checkpointing, etc.
- **Presentation**: data formats, transformation services.
- **Application**: interface between network and end-user programs

PHY



MAC



IP



Transport



APP

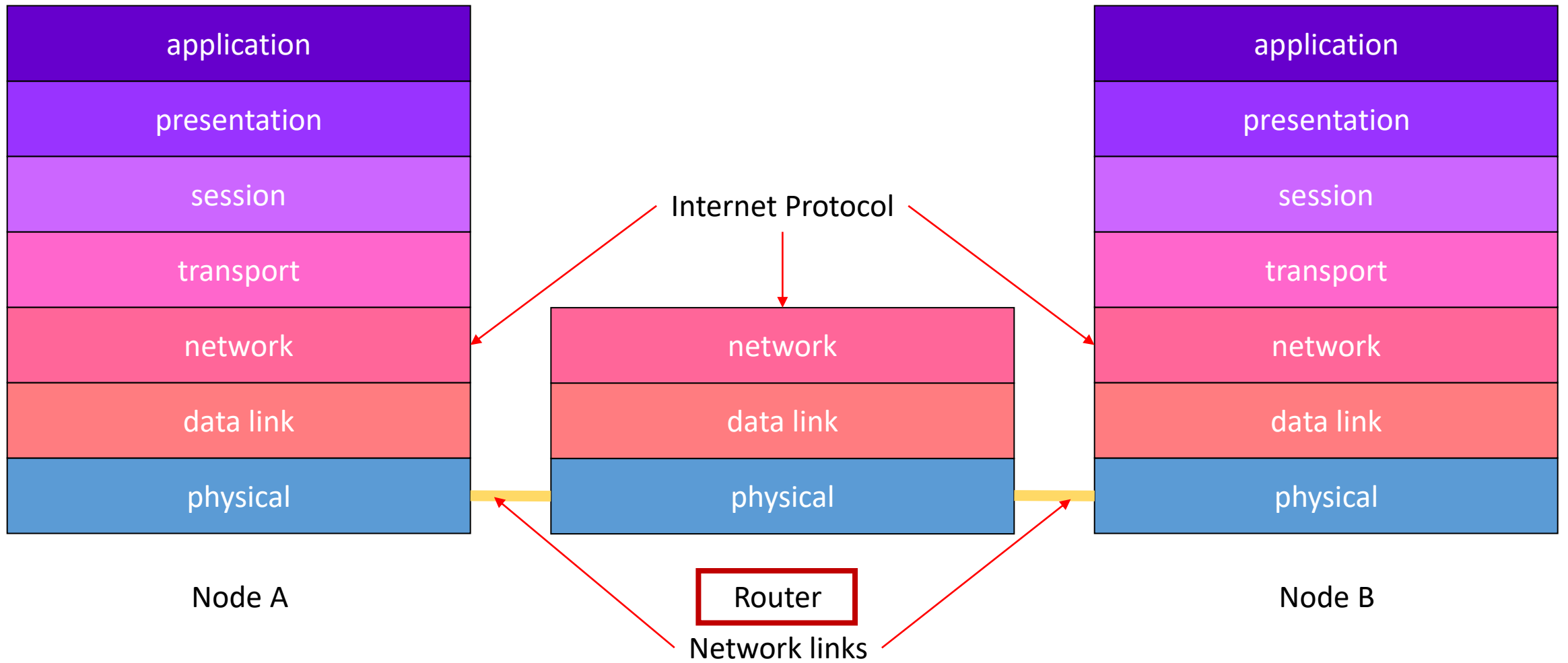


PHY and MAC

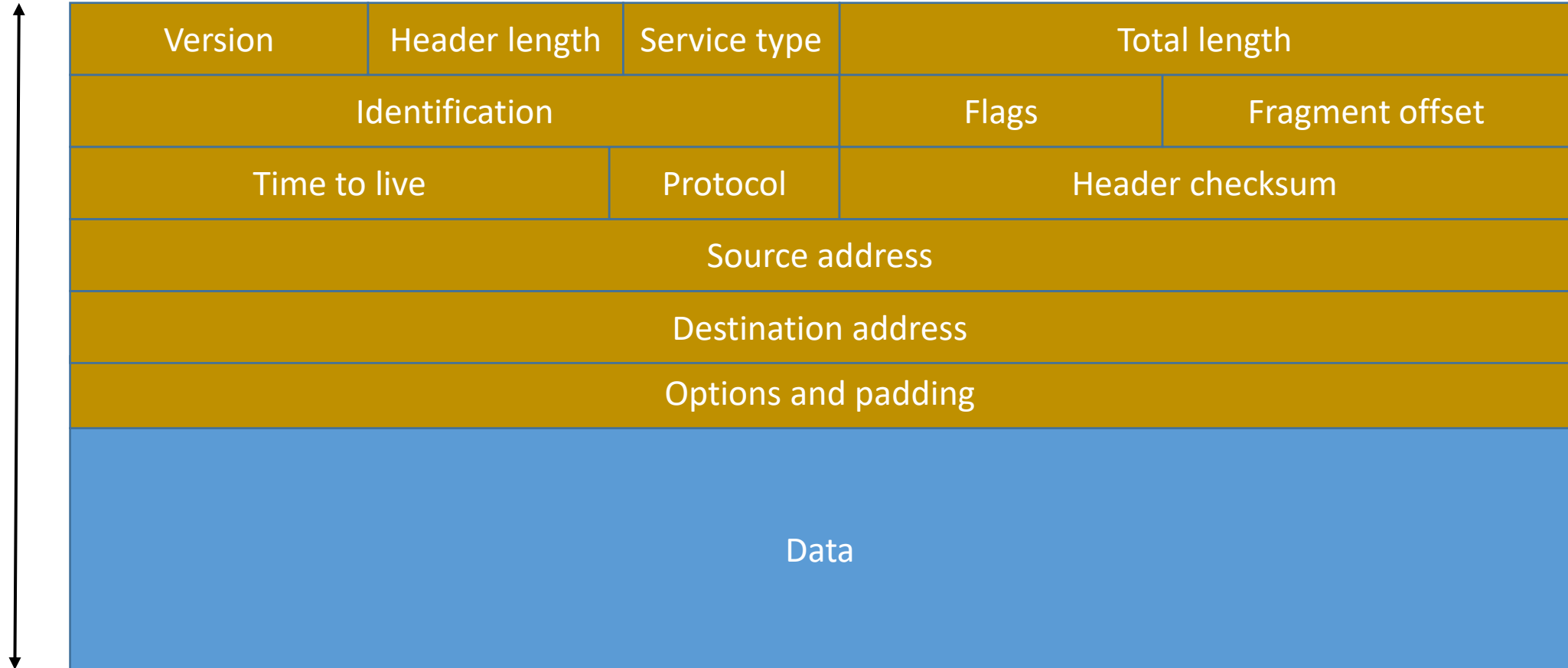
- PHY = physical layer.
 - Circuitry to transmit and receive bits.
- MAC = media access control.
 - Provides link-level services.

Internet Protocol

- Internet = network of networks.
 - Transports data from one network to another.
- The Internet uses Internet Protocol (IP).
 - Isolated networks can also use IP.



max 65,535 bytes



IP routing

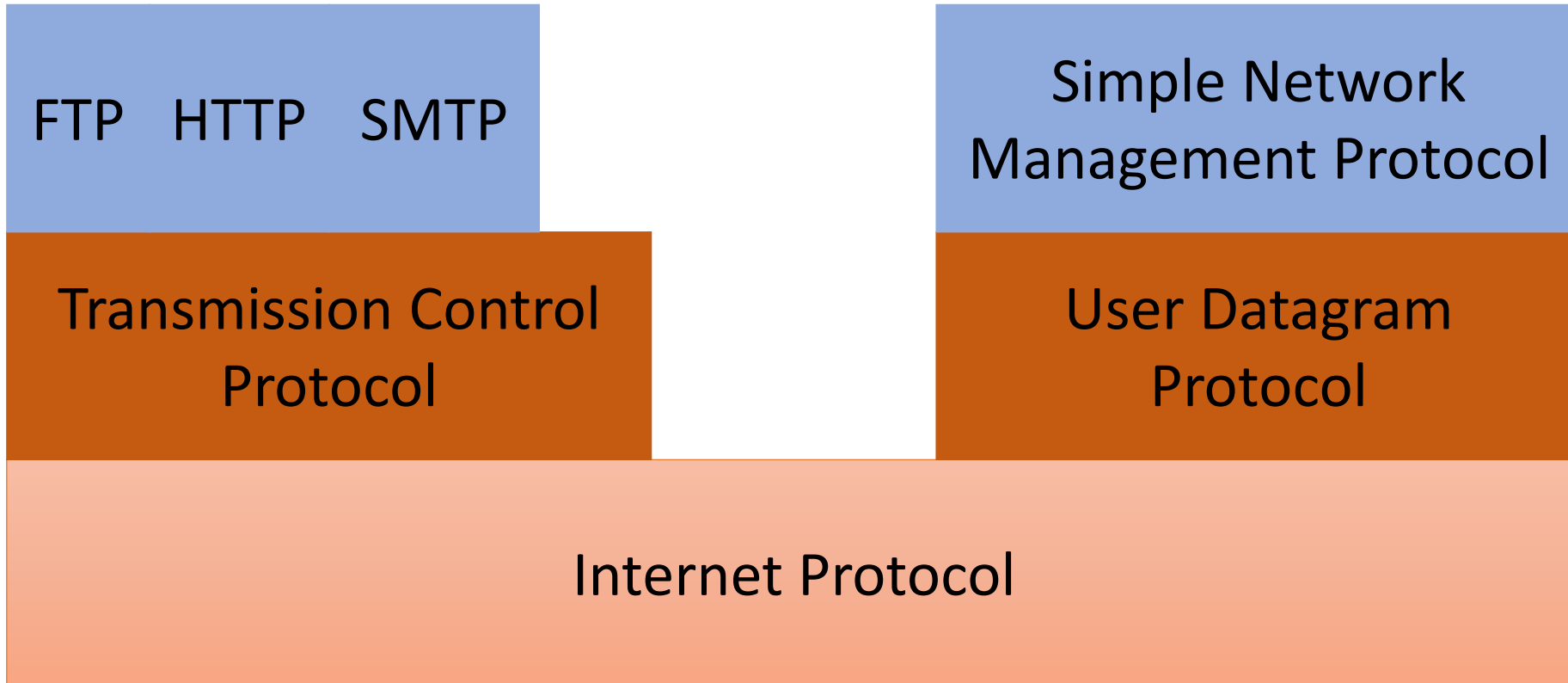
- IP routing is best effort---no guarantees of packet delivery.
- Build other services on top of IP:
 - Use handshakes to verify delivery of packet.
 - Network routers can enforce quality-of-service.

Internet services

- Domain name service (DNS):
 - Map names onto IP addresses.
- File transfer (FTP):
 - Move files from machine to machine.
- Terminal sessions:
 - Telnet provides terminal-style access.
- Web (HTTP):
 - Built on top of FTP.
- Email (SMTP):
 - Built on top of FTP.

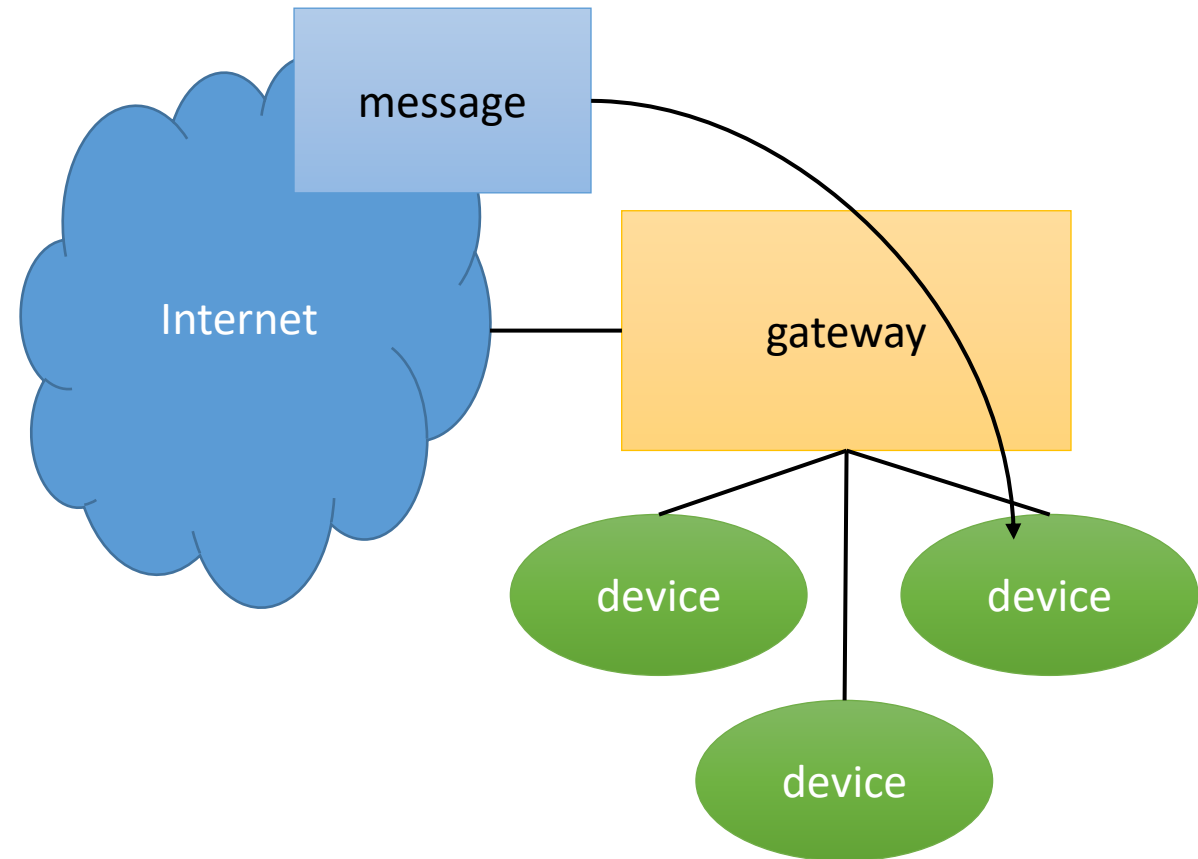
Host-to-host service

- Hides some details of IP:
 - Break host communication into IP packets at source.
 - Reassemble packets at destination.
 - Use handshake to ensure packets arrive, retransmit if necessary.
- Transmission Control Protocol (TCP):
 - Connection-oriented service.
- User Datagram Protocol (UDP):
 - Datagram service.
 - Datagram is modeled after telegram.

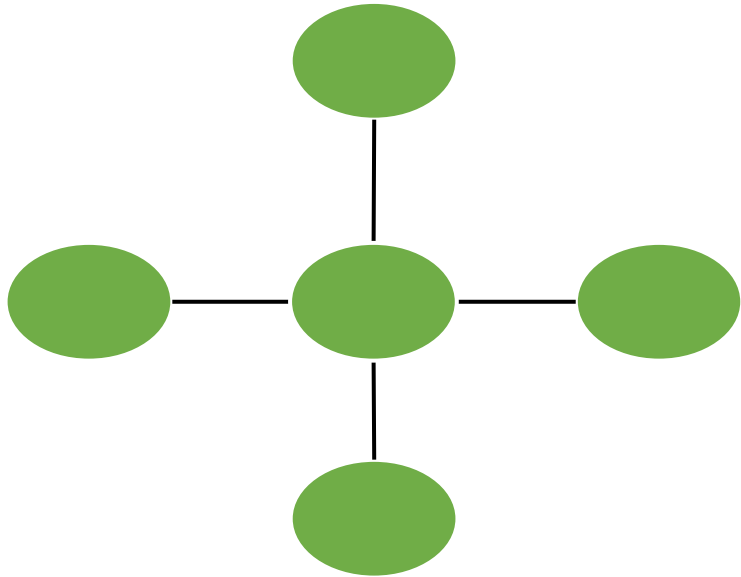


IoT networking concepts

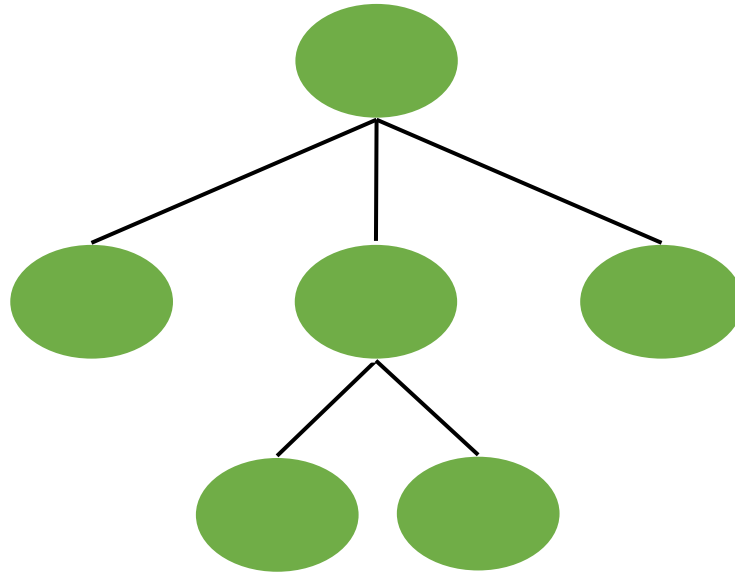
- Edge device may not run IP protocol.
 - IP connection may be provided by hub or gateway.
 - Non-IP networks are known as edge networks.
- Ad hoc network is self-organized---not set up by system administrator.
- Ad hoc network services:
 - Authentication of eligibility to join network.
 - Authorization for access to given pieces of information on the network.
 - Encryption and decryption.



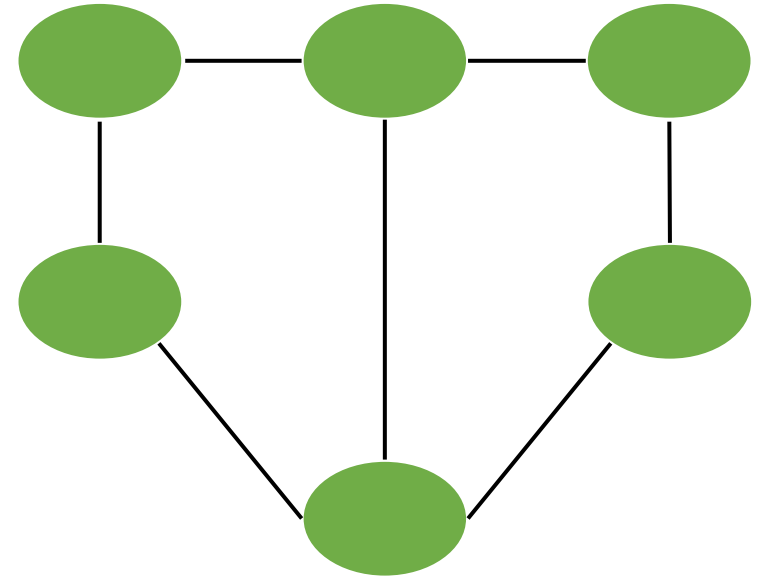
Network topologies



star



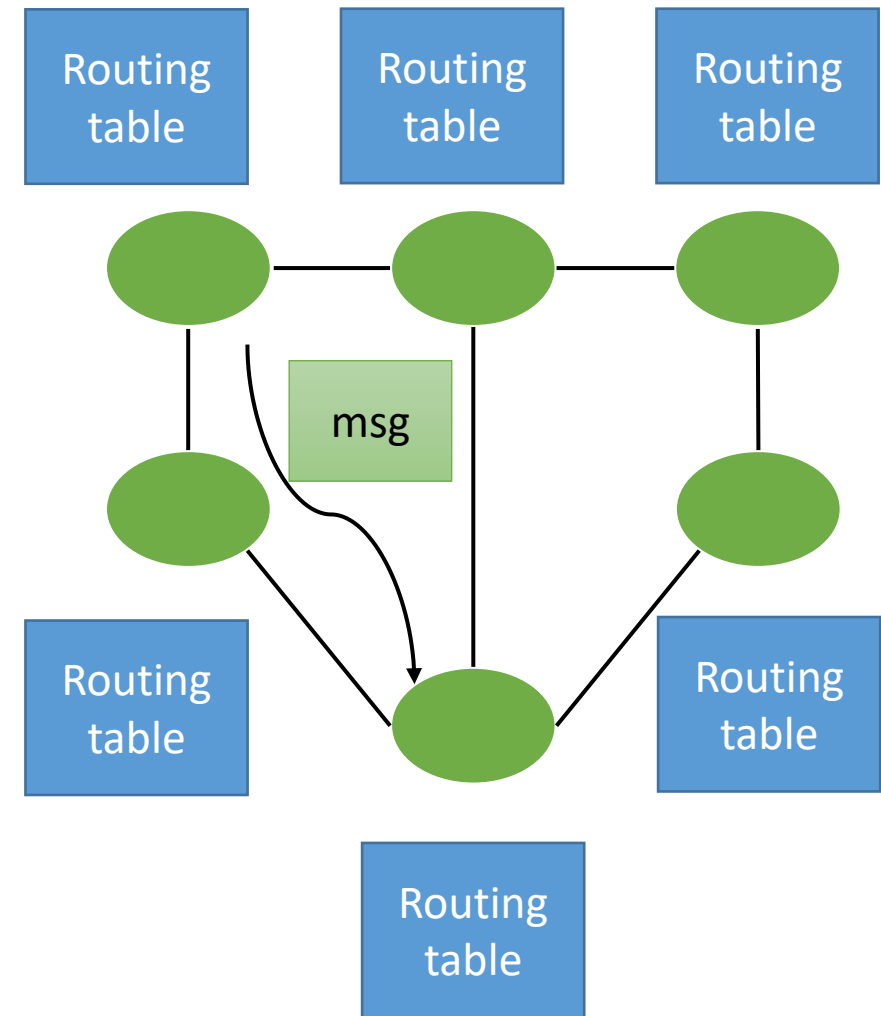
tree



mesh

Routing

- Routing discovery determines routes between source/destination pairs.
- Routing is driven by routing tables at the nodes.

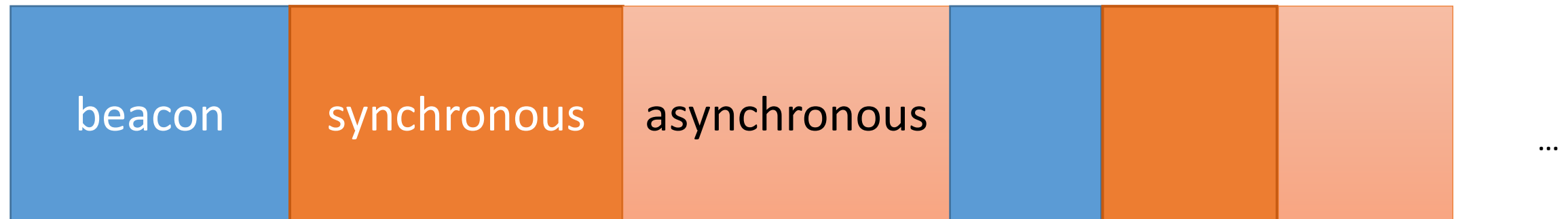


QoS

- Many networks support synchronous and asynchronous communication.
 - Asynchronous: data records, etc.
 - Synchronous: voice, etc.
- Quality-of-service (QoS): bandwidth and periodicity characteristics.
- Admission control ensures that network can handle the QoS demands of a request.

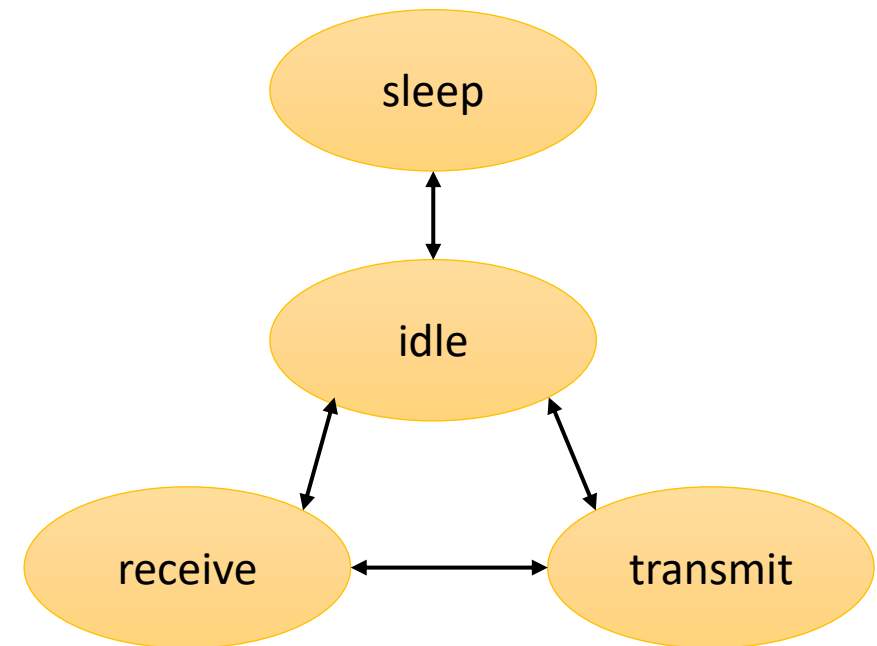
Synchronization and beacons

- Many network operations require nodes to be synchronized.
- Synchronization can be performed using beacon.
 - Beacon transmission marks the beginning of a communications interval.

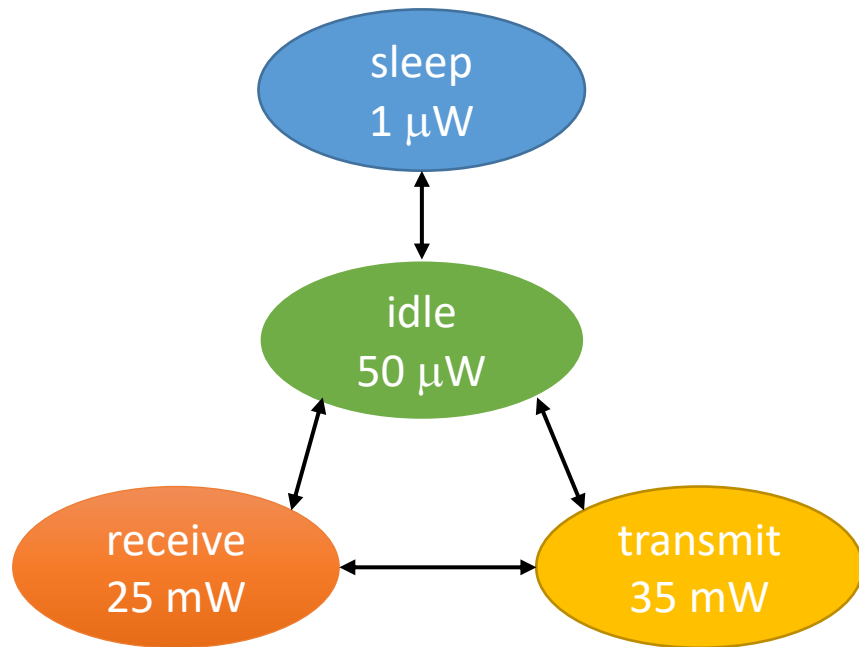


Communications energy

- Communications energy is a large part of node energy consumption.
- Comm energy consumption depends on many factors and parameters.
 - Generally evaluated for a set of use cases.
- We can use power state machine to model communications energy cost.



Communications power state machine example



step	state	time	energy
1	sleep	1 ms	1 nJ
2	idle	10 μs	0.5 nJ
3	receive	50 μs	1.25 μJ
4	transmit	50 μs	1.75 μJ
5	receive	50 μs	1.25 μJ
6	transmit	50 μs	1.75 μJ
			total = 6 μJ

Bluetooth

- Introduced in 1999, originally for telephony applications.
- Classic Bluetooth operates in instrumentation, scientific, and medical (ISM) band in the 2.4 GHz range.
- Bluetooth networks organized as piconet.
 - One master, several slaves.
 - Slave can be active or parked.
 - A device can be a slave on several networks simultaneously.

Bluetooth stack

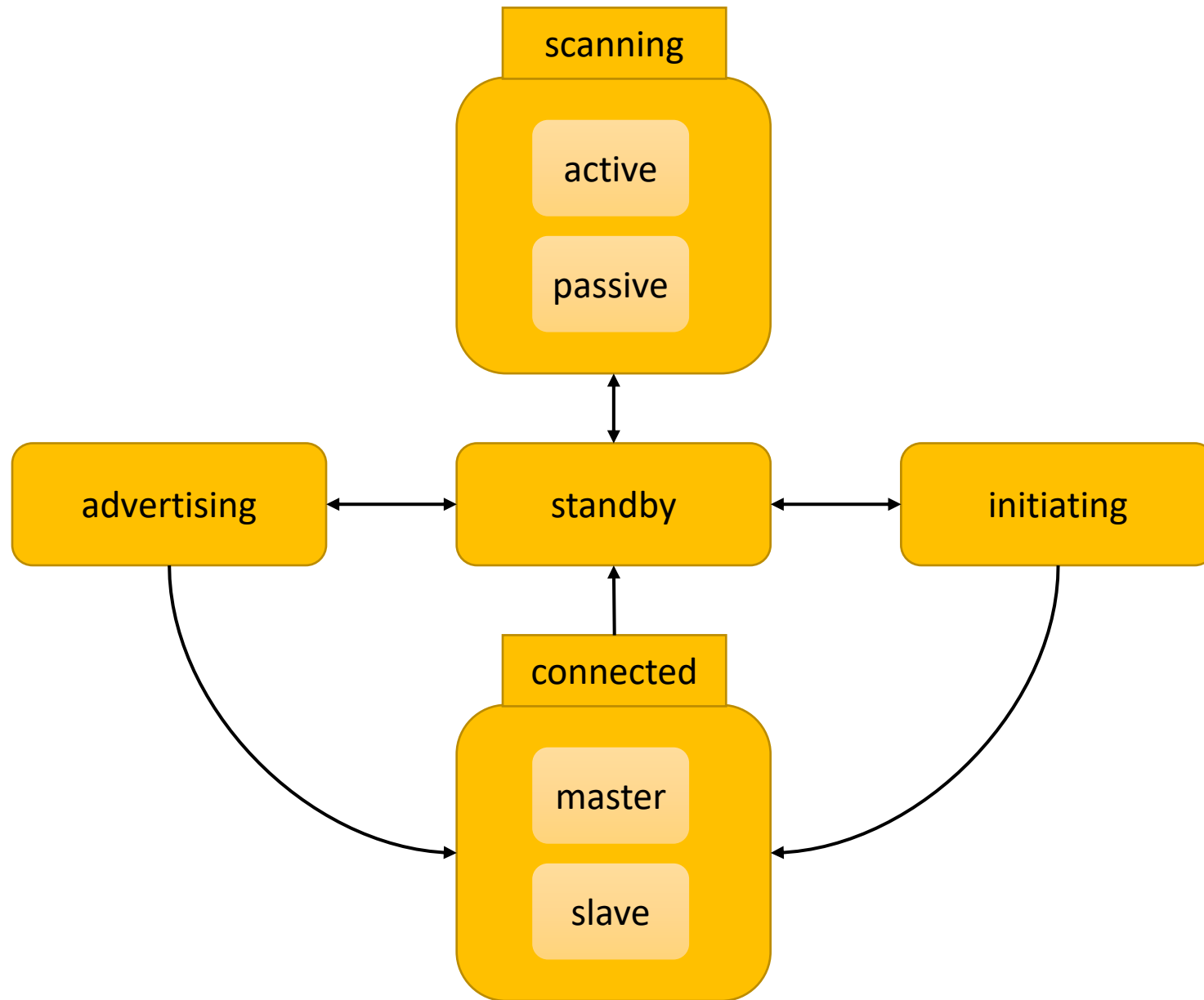
- Transport protocol:
 - Radio, baseband layer, link manager, logical link control and adaptation protocol (L2CAP).
- Middleware:
 - RFCOMM for serial port, service discovery protocol, Internet Protocol, IrDA, etc.
- Applications.

Bluetooth protocol

- Every Bluetooth device has a 48-bit Bluetooth Device Address.
- Every device has a Bluetooth clock.
- Transmissions alternate between master and slave directions.
- Two types of packets:
 - Synchronous connection-oriented (SCO) packets for QoS-oriented traffic.
 - Asynchronous connectionless (ACL) packets for non-QoS traffic.
 - SCO traffic has higher priority than ACL packets.

Bluetooth Low Energy

- Designed for very low energy operation such as button-sized battery.
 - Goal: minimize radio on-time.
- Part of Bluetooth standard but deviates from Classic Bluetooth in several ways.
- Advertising transmissions can be used to broadcast, discover devices, etc.
- Connections can be established.
- Attribute Protocol Layer allows devices to create application-specific protocols.
- Generic Attribute Profile Layer (GATT) defines basic attributes for all BLUE devices.
- Pairing devices uses a short-term key to send a long-term key.
 - Bonding: storing long-term key in device database.
 - Optional data encryption using AES.



802.15.4 and ZigBee

- 802.15.4 defines MAC and PHY layers.
 - Supports full-function and reduced-function devices.
 - Either star or peer-to-peer topology.
 - Communication performed using frames.
 - Optional superframe provides a beacon mechanism and QoS.
- ZigBee is a set of application-oriented standards.
 - NWK layer provides network services.
 - APL layer provides application-level services.
 - Supports many different topologies.

Wi-Fi

- Originally designed for portable and mobile applications.
 - Has been adapted for lower-energy operation.
- Supports ad hoc networking.
- Network provides a set of services:
 - Distribution of messages from one node to another.
 - Integration delivers messages from another network.
 - Association relates a station to an access point.

IoT systems

- Databases.
- Timewheels.
- Example: smart home.

Databases

- Database holds data about devices, helps to analyze data.
- Relational database management system:
 - Domain1 X domain2 X ... -> Range.
- Database organized into records or tuples:
 - Attribute: table column.
 - Record: table row.
- One column is the primary key---uniquely identifies a record.

Database example

devices

record

name	id (primary key)	address	type
door	234	10.113	binary
refrigerator	4326	10.117	signal
table	213	11.039	MV
chair	4325	09.423	binary
faucet	2	11.324	signal

device_data

signature (primary key)	device	time	value
256423	234	11:23:14	1
252456	4326	11:23:47	40
663443	234	11:27:55	0

Schemas

- Schema: definition of tables in database.
- Eliminating redundant storage of data helps to ensure database consistency.
- Normal forms help eliminate redundancies and other problems:
 - First normal form: every cell contains a single value.
 - Second: First + values of cells in a record are unique to the key.
 - Third: Second + non-key columns are independent.

Queries

- Query: request for information from database.
- Structured query language (SQL) isolates user from details of schema.
 - Select from device_data where device = 234
- Join: combine information from more than one table.
- Projection: eliminate some columns.
- Restriction: eliminate some rows.

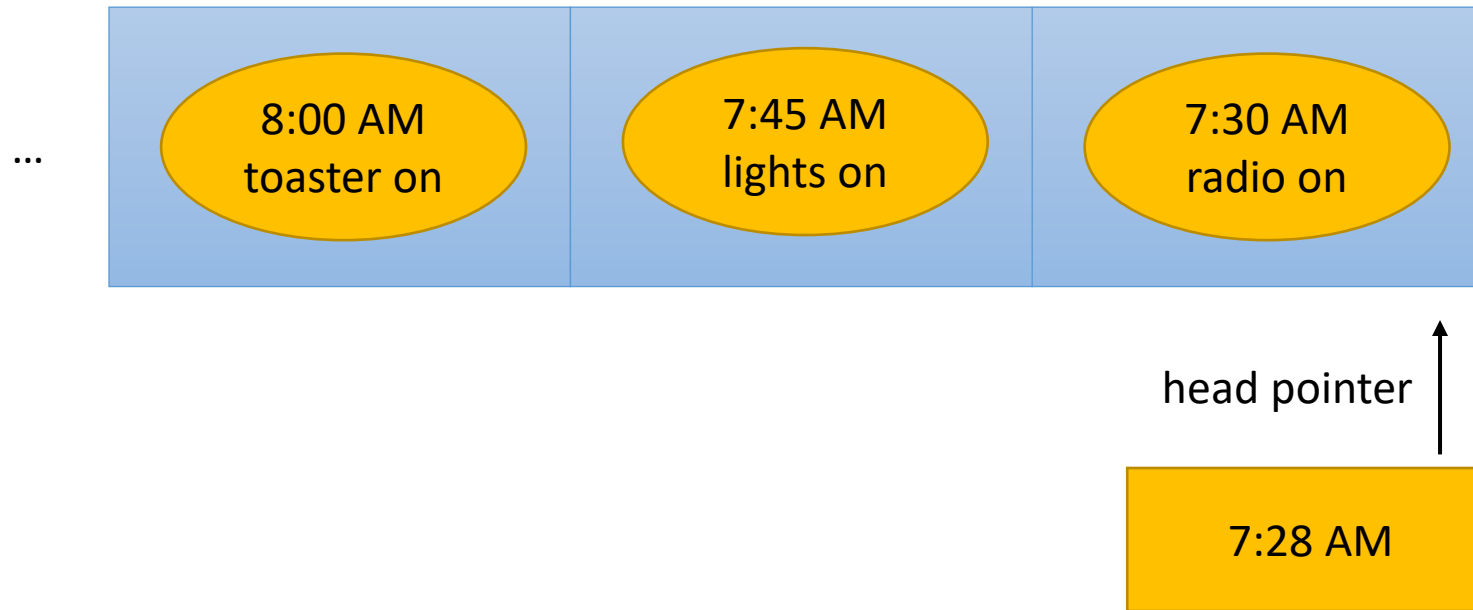
Schemaless databases

- Some simple data models do not have a schema.
 - Schemaless or noSQL.
- JSON is an example schemaless language.
 - Name/value pairs, ordered lists of values.
- Schemaless databases may be simpler to set up, harder to maintain.

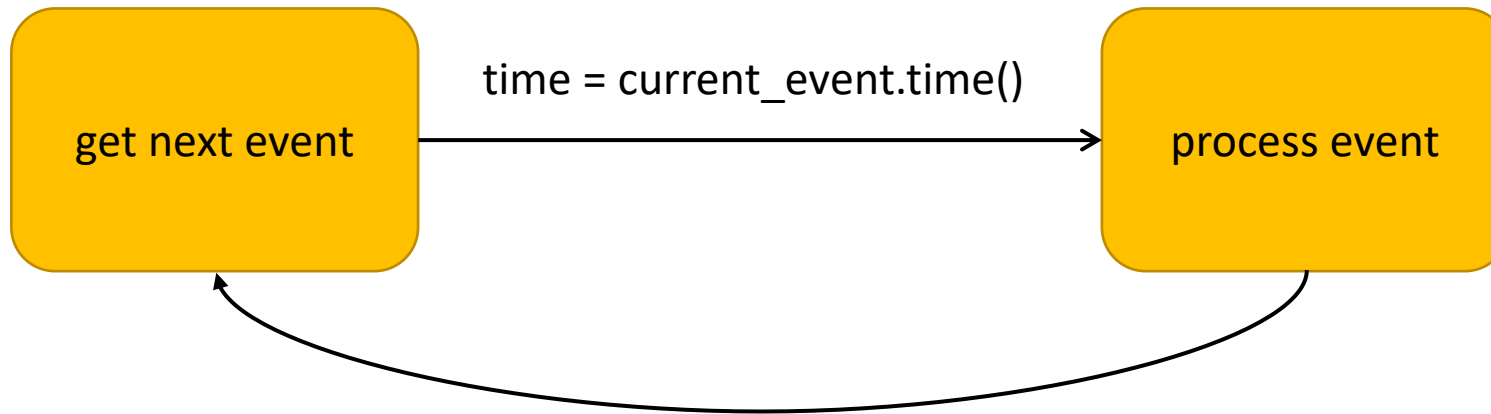
Timewheels

- Used to manage timing of events in the system.
- Timewheel is a time-sorted set of events.
 - Event placed in proper spot in timewheel queue upon arrival.
 - When current time is equal to time of event at head, event is processed.

Timewheel data structure



Timewheel state diagram

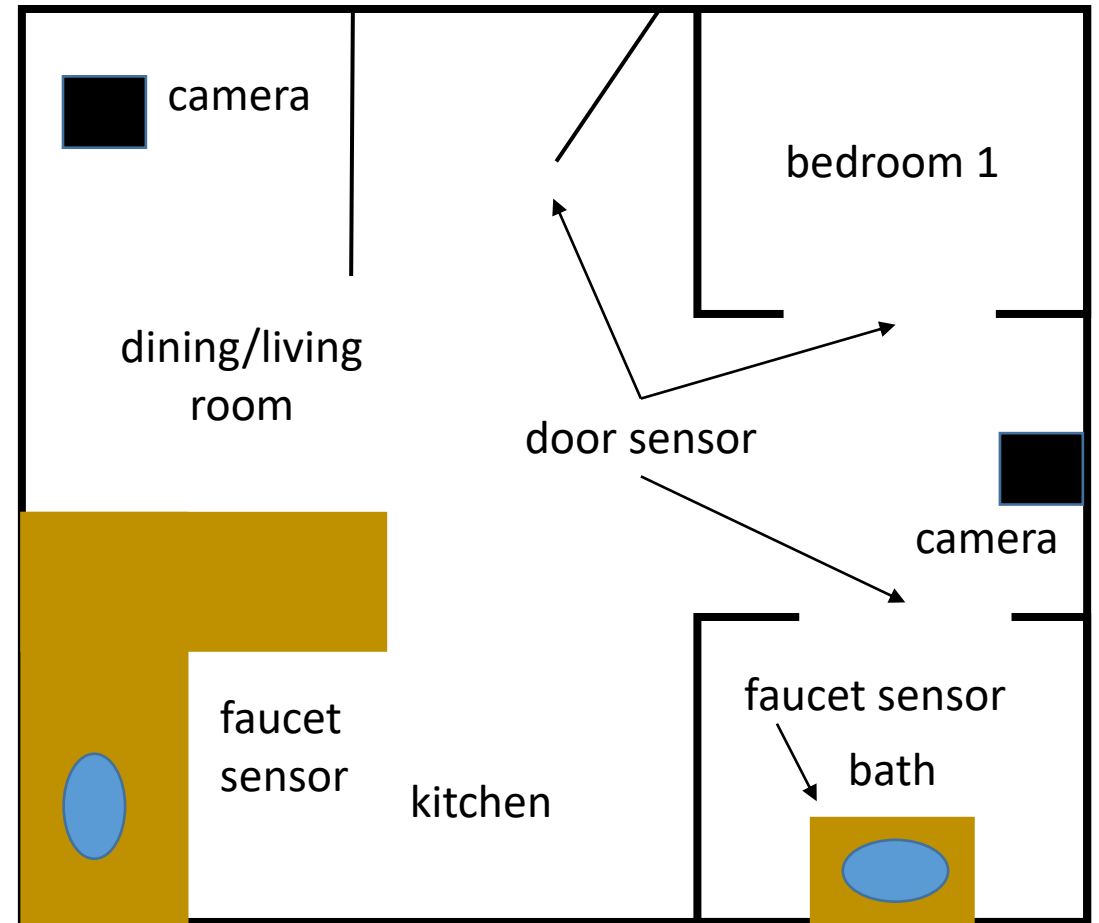


Example: smart home

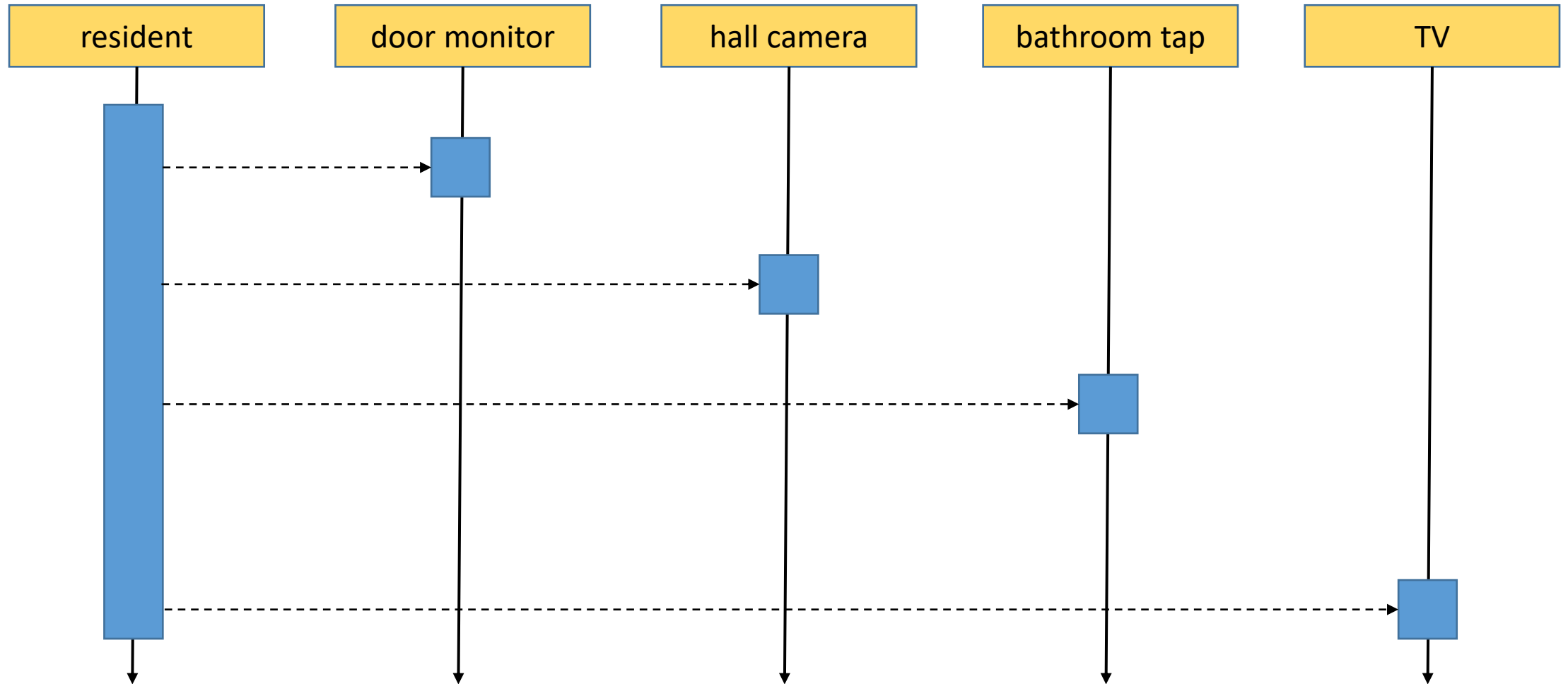
- Performs a variety of services:
 - Remote or automatic operation of lights and appliances.
 - Energy and water management.
 - Activity monitoring.
- Activity monitoring can help elderly, people with special needs:
 - Reports on daily activities.
 - Alerts for out-of-the-ordinary activity.
 - Recommendations.

Example smart home

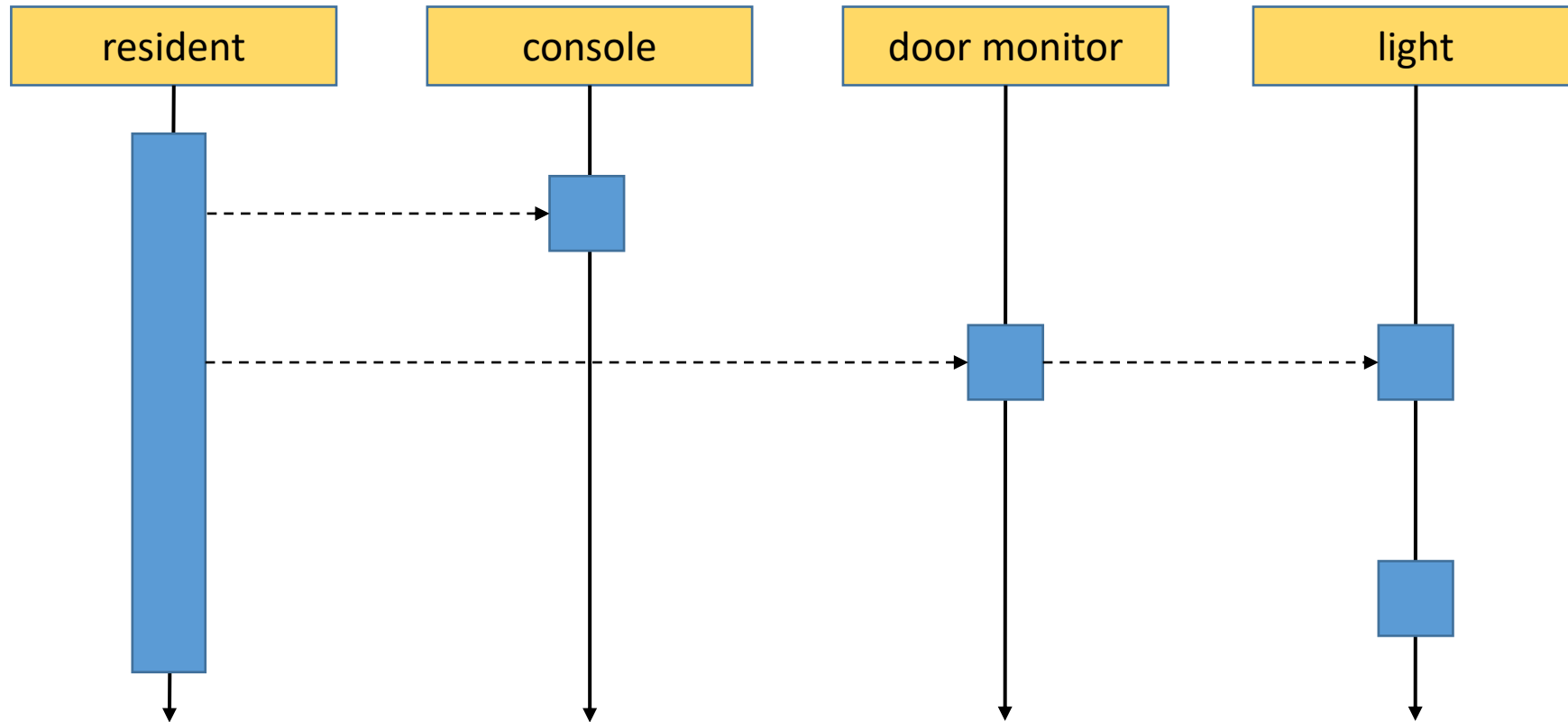
- Cameras can identify resident and their activity.
- Faucet, door sensors can identify activity but not who performs the activity.



Use case: activity monitoring



Use case: light control



Smart home object diagram

